Bench Drill Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Bench Drill Machine (With belt safety guard & on/off switch facility)

What it is used for

A bench drill machine uses a rotating drill bit to make controlled holes in suitable workpieces secured on a worktable. Correct bit selection, secure clamping, guarded belt operation, and controlled feed help improve drilling accuracy and reduce entanglement, impact, and cutting injuries.

Main parts

- Base supports the machine and worktable assembly.
- Column supports the drill head and table.
- Drill head houses the spindle and drive mechanism.
- Electric motor powers the spindle through the drive system.
- Belt and pulleys transmit drive and provide speed selection.
- Belt safety guard covers moving belts and pulleys.
- Chuck holds the drill bit.
- Feed handle lowers and raises the spindle.
- Worktable supports the workpiece and may be adjustable.
- On/off switch controls electrical operation.

Before you start

1. Only trained personnel should operate the bench drill machine.
2. Inspect the machine, belt guard, chuck, drill bit, switch, power cable, and table for damage.
3. Confirm the drill bit is sharp, correctly fitted, and suitable for the material and hole size.
4. Ensure the belt guard is closed and secured before switching on.
5. Select the correct spindle speed according to the material, bit diameter, and manufacturer's guidance; change belt positions only with power isolated.
6. Clamp the workpiece securely to the table or suitable machine vice; never hold it by hand while drilling.
7. Remove the chuck key before starting and wear eye protection and other task-appropriate PPE.

How to use

1. Secure the machine on a stable bench and ensure the work area is clear and well lit.
2. Isolate electrical power before fitting the drill bit or adjusting the belt and pulleys.
3. Insert the correct drill bit into the chuck and tighten it securely using the approved method.
4. Remove the chuck key and check that the belt guard is properly closed and secured.
5. Adjust the worktable height and position, then clamp the workpiece securely.

6. Set the appropriate spindle speed and verify that all adjustments are locked.
7. Connect power and switch on the machine while standing clear of the rotating spindle.
8. Check for abnormal vibration, noise, or runout before contacting the workpiece.
9. Lower the drill bit gradually using the feed handle and apply steady pressure appropriate to the material.
10. Withdraw the bit periodically when needed to clear chips, using a safe procedure and keeping hands away from the rotating bit.
11. Switch off the machine and wait for the spindle to stop completely before measuring or handling the workpiece.
12. Remove chips with a brush or suitable tool after the machine has stopped; never clear them by hand.
13. Isolate power before adjustments, bit changes, cleaning, or maintenance.

⚠ Safety

- Never operate the machine with the belt safety guard open, removed, or damaged.
- Tie back long hair and secure loose clothing, jewellery, and other items that could become entangled.
- Do not wear gloves near rotating drill bits or spindles.
- Never hold a workpiece by hand or attempt to stop a rotating chuck by hand.
- Keep hands away from the rotating bit and remove the chuck key before starting.
- Disconnect and isolate power before changing belts, adjusting pulleys, or servicing the machine.

Care & storage

- Clean chips and dust from the table and machine after isolating power.
- Inspect the drill bit, chuck, belt, pulleys, guard, and switch regularly.
- Lubricate and maintain moving parts only as specified by the manufacturer.
- Keep the table and machine surfaces clean, dry, and free from loose tools.
- Report unusual noise, vibration, belt damage, or switch faults and arrange authorised maintenance.

Common mistakes to avoid

- Starting the machine with the chuck key still inserted.
- Drilling an unsecured workpiece or using an incorrect spindle speed.
- Opening the belt guard or changing belt positions while power is connected.
- Wearing loose clothing or gloves near rotating parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.